



TRINITY GRAMMAR SCHOOL

Mathematics Department

--	--	--	--	--	--	--	--	--

(NESA Student Number | Year 12 only)

2024

Year 12

Mathematics Extension 1

TRIAL EXAMINATION

ASSESSMENT TASK 4

Date of Assessment Task: Friday, 30 August 2024

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen.
- Erasable pens and all electronic devices (or timekeepers), unless otherwise advised, are not permitted in this Assessment.
- Calculators approved by NESA may be used.
- A reference sheet is provided.
- For questions in Section II, show relevant mathematical reasoning and/or calculations.
- *Write your NESA Student Number (Year 12) at the top of this page*

Total marks:
70

Section I – 10 marks (pages 2 – 6)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II – 60 marks (pages 7 – 13)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section.

- Year 12 HSC Assessment Weighting: 30%

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1–10.

1 Which of the following vectors are perpendicular?

A. $\underline{u} = 3\underline{i} - 7\underline{j}, \underline{v} = \frac{7}{3}\underline{i} - \underline{j}$

B. $\underline{u} = \frac{2}{5}\underline{i} - 2\underline{j}, \underline{v} = \frac{7}{3}\underline{i} - \underline{j}$

C. $\underline{u} = \frac{9}{2}\underline{i} + \frac{7}{2}\underline{j}, \underline{v} = -\frac{2}{9}\underline{i} - \frac{1}{7}\underline{j}$

D. $\underline{u} = 14\underline{i} + \underline{j}, \underline{v} = \frac{1}{2}\underline{i} - 7\underline{j}$

2 What is the solution to the inequality $|2 - 5x| < 7$?

A. $x < -1$ or $x > \frac{9}{5}$

B. $-1 < x < \frac{9}{5}$

C. $x < -\frac{9}{5}$ or $x > 1$

D. $-\frac{9}{5} < x < 1$

3 What is the value of $\sin 2x$ given that $\sin x = \frac{\sqrt{3}}{4}$ given x is obtuse angled?

A. $-\frac{\sqrt{39}}{8}$

B. $-\frac{\sqrt{3}}{2}$

C. $\frac{\sqrt{3}}{2}$

D. $\frac{\sqrt{39}}{8}$

- 4 The radius of a spherical balloon is expanding at a constant rate of 1.3 cm s^{-1} . What is the rate of change of the surface area of the balloon when its radius is 6.3 cm?

A. $68.61 \text{ cm}^2 \text{ s}^{-1}$
B. $158.34 \text{ cm}^2 \text{ s}^{-1}$
C. $205.84 \text{ cm}^2 \text{ s}^{-1}$
D. $498.76 \text{ cm}^2 \text{ s}^{-1}$

- 5 Using the digits 0, 1, 2, 3, 4 and 5, how many odd numbers between 2000 and 5000 can be formed if no digits are repeated?

A. 48
B. 96
C. 120
D. 360

- 6 Consider the integral:

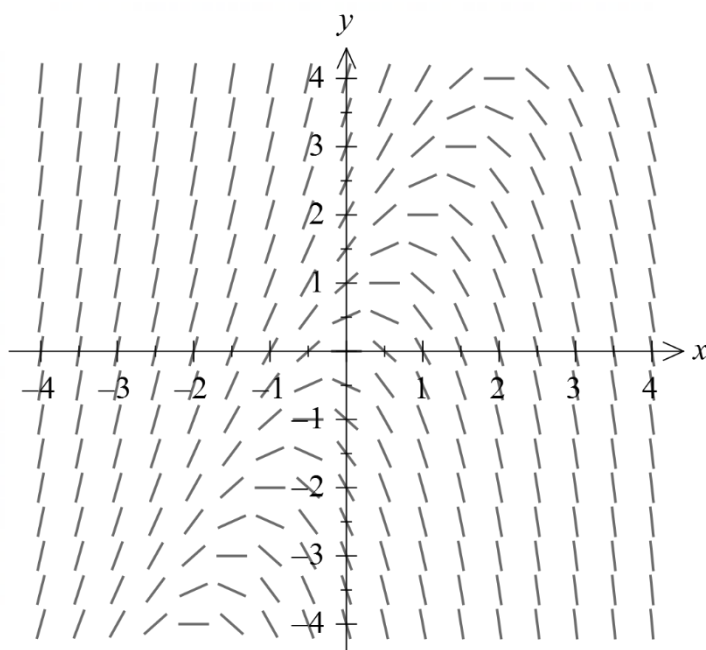
$$\int_0^k \frac{2}{8 + (2x)^2} dx = \frac{\pi}{8\sqrt{2}}$$

What is the value of k ?

A. $\frac{1}{\sqrt{3}}$
B. 1
C. $\sqrt{2}$
D. $\sqrt{3}$

- 7 The polynomial $x^2 + x + 2$ is a factor of the polynomial $x^4 + 4x^2 + x + a$.
What is the value of a ?
- A. -6
B. -3
C. 3
D. 6

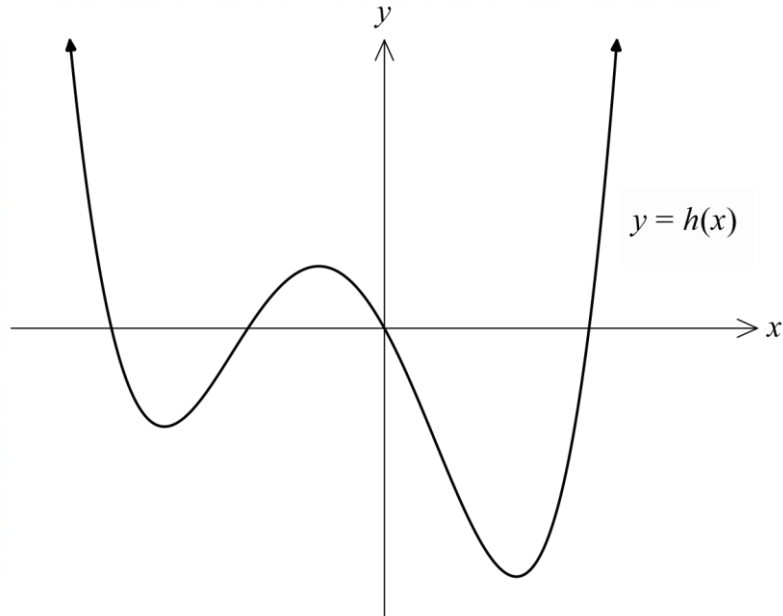
- 8 The direction field for a differential equation is shown below.



Which of the following equations could give this direction field?

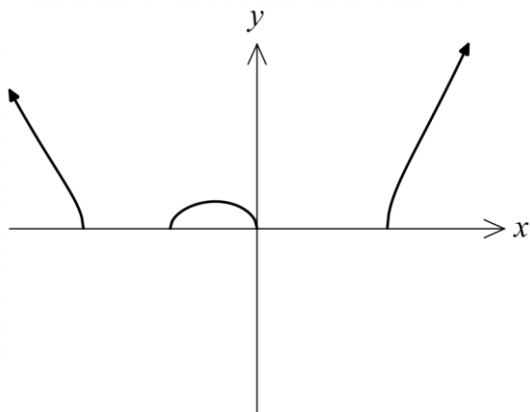
- A. $y' = y - 2x$
B. $y' = y + 2x$
C. $y' = y + x$
D. $y' = x - y$

- 9 The graph of $y = h(x)$ is shown below.

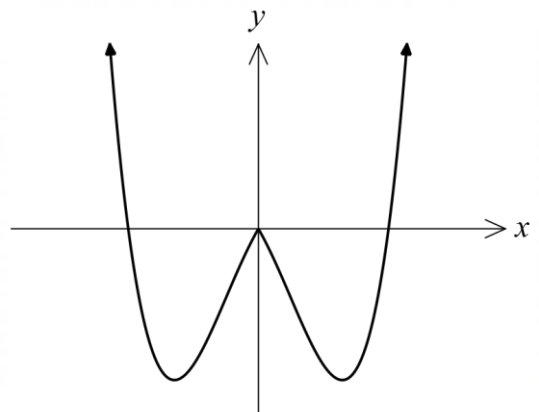


Which graph below shows $y = |h(x)|$?

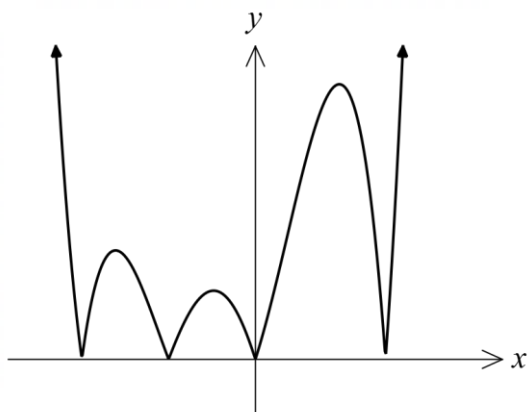
A.



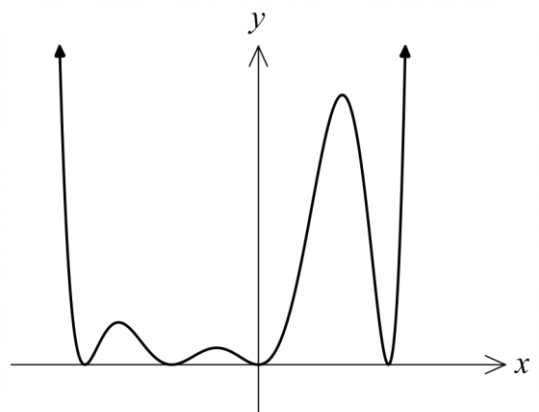
B.



C.



D.



- 10** The graph of the function $y = \sin^{-1}\left(\frac{1}{2}(x-2)\right)$ is to be transformed by a translation by 1 unit to the left, then a horizontal dilation with a scale factor of 2.

What is the domain of the resultant function?

- A. $-1 \leq x \leq 7$
- B. $-2 \leq x \leq 6$
- C. $1 \leq x \leq 3$
- D. $-\frac{1}{2} \leq x \leq \frac{3}{2}$

Section II

60 marks

Attempt Questions 11-14

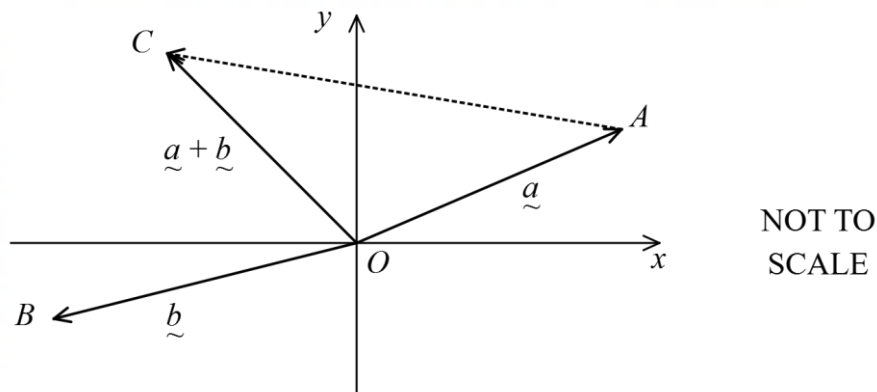
Allow about 1 hour and 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra Writing booklets are available.

In Questions 11-14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (16 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows the vectors $\underline{a} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} -9 \\ -1 \end{pmatrix}$ and the resultant vector $\underline{a} + \underline{b}$.



- (i) Show that $\triangle AOC$ is an isosceles triangle. 2
- (ii) Find the size of $\angle OAC$. 1

Question 11 continues on page 8

Question 11 (continued)

(b) (i) Rewrite $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$, $R > 0$. 2

(ii) Hence, or otherwise, solve $\cos x - \sqrt{3} \sin x = 1$ for $0 \leq x \leq 2\pi$. 2

(c) Evaluate $\int_1^{\sqrt{2}} \frac{x}{\sqrt{4-x^2}} dx$ using the substitution $u = 4 - x^2$. 3

(d) The coefficient of x in the expansion of $\left(x + \frac{1}{ax^2}\right)^7$ is $\frac{7}{3}$. 3
Find the value(s) of a .

(e) Solve the inequality $\frac{1}{x} \leq \frac{1}{x+1}$. 3

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Use mathematical induction to show that $9^{n+2} - 4^n$ is divisible by 5 for all positive integers n . 4
- (b) The polynomial $P(x) = x^4 - 3x^3 + ax^2 + bx - 6$ leaves a remainder of 8 when divided by $(x + 1)$. If $(x - 3)$ is a factor of $P(x)$, find the values of a and b . 4
- (c) Consider the function $f(x) = \sqrt{4 - x^2}$. A new function is created such that $g(x) = \frac{1}{f(x)}$.
- (i) Sketch the graph of $y = g(x)$, showing all important features. 2
(You do NOT need to use calculus to verify any points on the graph.)
- (ii) Find the exact area bounded by $y = g(x)$, $x = \sqrt{2}$, $x = \sqrt{3}$ and the x -axis. 3
- (d) Given that ${}^nC_4 = p$ and ${}^nC_5 = q$, use the identities of Pascal's Triangle, or otherwise, to find a simplified expression for ${}^{n+1}C_{n-5} + {}^nC_{n-4}$ in terms p and q . 2

End of Question 12

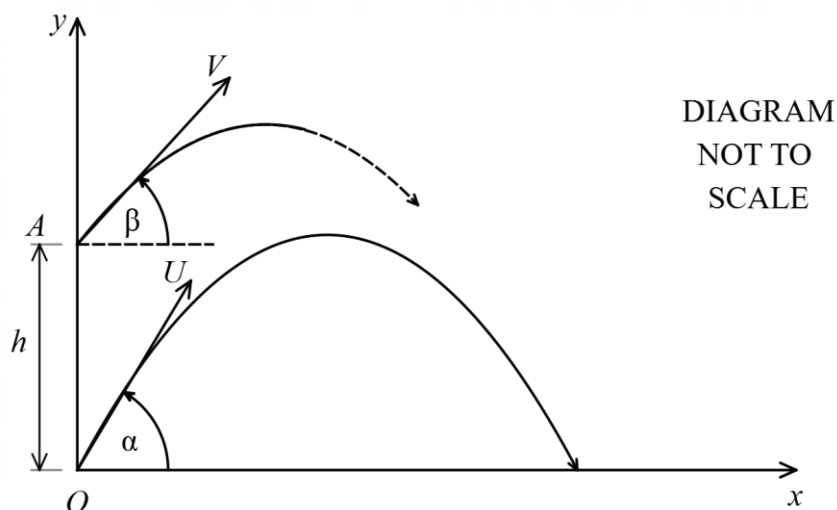
Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The letters of the word PERSEVERE are arranged in a row.
- (i) How many different arrangements of the letters are possible? **1**
- (ii) What is the probability that an arrangement will have all E's together in one group and all the R's together in one group? **2**
- (b) If $2\sin\left(\theta - \frac{\pi}{3}\right) = \cos\left(\theta - \frac{\pi}{3}\right)$ express $\tan\theta$ in the form $a + b\sqrt{c}$ where a , b and c are integers. **3**
- (c) Consider $f(x) = 4\sin^2\left[2\left(x + \frac{\pi}{8}\right)\right] - 2$ for $0 < x < \frac{\pi}{4}$.
- (i) Verify that $f(x)$ could be a continuous probability density function. **3**
- (ii) Find the mode of the distribution. **2**
- (d) The curve $y = -\log_e(x + 1)$, the y -axis and the line $y = 1$ enclose a region in the number plane.
- (i) Sketch the region, clearly showing any asymptotes and intercepts with axes. **1**
- (ii) The region is rotated about the y -axis to form a solid of revolution. Find the volume, giving your answer correct to 3 significant figures. **3**

End of Question 13

Question 14 (14 marks) Use a SEPARATE writing booklet.

- (a) An object is projected on level ground from the origin, O , with initial speed U m/s at an angle of elevation of α . At the same instant another object is projected from a point A which is h m above the origin. The second object is projected with initial speed V m/s at an angle of elevation of β , where $\beta < \alpha$.



- (i) The parametric equations of displacement for the object projected from the origin, O , are **1**

$$x = U t \cos \alpha \text{ and } y = U t \sin \alpha - \frac{1}{2} g t^2$$

(DO NOT PROVE THESE)

Write down the two parametric equations of displacement for the object projected from A .

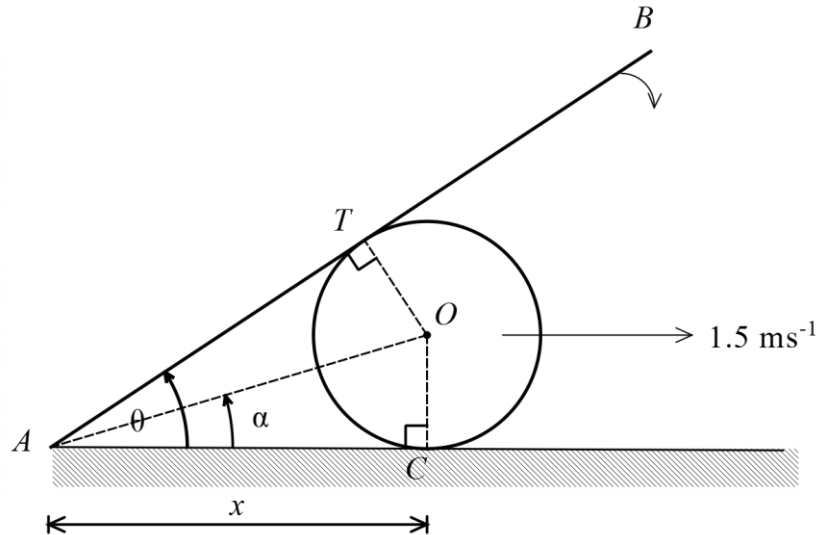
- (ii) If the two objects collide T seconds after they are projected, show that **4**
- $$T = \frac{h \cos \beta}{U \sin(\alpha - \beta)}.$$

Question 14 continues on page 12

Question 14 (continued)

- (b) A rod AB is leaning on a metal disc of radius 1 m, as shown in the diagram below. The centre of the disc is at O .

One end of the rod, A , is fixed to the ground, while the disc is tangential to the ground at C , such that $\angle ACO = \angle ATO = 90^\circ$.



The disc rolls horizontally to the right with a constant speed of 1.5 ms^{-1} . As the disc rolls the end of the rod in the air, B , descends to the ground.

Let $\angle OAC = \alpha$, $\angle BAC = \theta$ and $AC = x$.

- (i) Write an expression for $\tan \alpha$ in terms of x . 1
- (ii) Hence, or otherwise, find the rate of change of θ in radians per second when $x = 4 \text{ m}$, giving your answer correct 3 decimal places. 4

Question 14 continues on page 13

Question 14 (continued)

- (c) A tiny pond is initially unpolluted and holds 450 litres of water. There is an input stream with a concentration of 5 g/L of pollutant flowing into the pond at a rate of 30 L/day. Moreover, there is an output stream with a volumetric flow rate of 30 L/day.

Let m be the mass (in grams) of pollutant in the pond and t be the number of days after the input stream first begins to flow into the pond.

- (i) Show that a differential equation to model the mass of pollutant after t days is **1**
given by $\frac{dm}{dt} = 150 - \frac{m}{15}$.
- (ii) What is the concentration of the pollutant, m , in the pond after 10 days? **3**

End of Question 14

End of Paper



TRINITY GRAMMAR SCHOOL

Mathematics Department

--	--	--	--	--	--	--	--	--

(NESA Student Number | Year 12 only)

2024

SOLUTIONS

Year 12

Mathematics Extension 1

TRIAL EXAMINATION

ASSESSMENT TASK 4

Date of Assessment Task: Friday, 30 August 2024

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen.
- Erasable pens and all electronic devices (or timekeepers), unless otherwise advised, are not permitted in this Assessment.
- Calculators approved by NESA may be used.
- A reference sheet is provided at the back of this paper.
- For questions in Section II, show relevant mathematical reasoning and/or calculations.
- *Write your NESA Student Number (Year 12) at the top of this page!*

Total marks:
40

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II – 60 marks (pages 6 – 12)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section.

- **Year 12 HSC Assessment Weighting: 30%**

- *Shade completely only ONE answer for each question.*

1	☐ A	☐ B	☐ C	☒ D
2	☐ A	☒ B	☐ C	☐ D
3	☒ A	☐ B	☐ C	☐ D
4	☐ A	☐ B	☒ C	☐ D
5	☐ A	☒ B	☐ C	☐ D
6	☐ A	☐ B	☒ C	☐ D
7	☐ A	☐ B	☐ C	☒ D
8	☒ A	☐ B	☐ C	☐ D
9	☐ A	☐ B	☒ C	☐ D
10	☐ A	☒ B	☐ C	☐ D

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Which of the following vectors are perpendicular?

$$\underline{u} \cdot \underline{v} = 0$$

- A. $\underline{u} = 3\underline{i} - 7\underline{j}$, $\underline{v} = \frac{7}{3}\underline{i} - \underline{j}$
- B. $\underline{u} = \frac{2}{5}\underline{i} - 2\underline{j}$, $\underline{v} = \frac{7}{3}\underline{i} - \underline{j}$
- C. $\underline{u} = \frac{9}{2}\underline{i} + \frac{7}{2}\underline{j}$, $\underline{v} = -\frac{2}{9}\underline{i} - \frac{1}{7}\underline{j}$
- ☒ D. $\underline{u} = 14\underline{i} + \underline{j}$, $\underline{v} = \frac{1}{2}\underline{i} - 7\underline{j}$

- 2 What is the solution to the inequality $|2 - 5x| < 7$?

A. $x < -1$ or $x > \frac{9}{5}$

☒ B. $-1 < x < \frac{9}{5}$

C. $x < -\frac{9}{5}$ or $x > 1$

D. $-\frac{9}{5} < x < 1$

$$\begin{aligned} 2 - 5x &> -7 & \text{or} & & 2 - 5x < 7 \\ -5x &> -9 & & & -5x < 5 \\ x &< \frac{9}{5} & & & x > -1 \end{aligned}$$

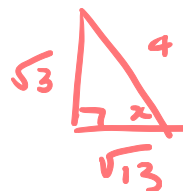
- 3 What is the value of $\sin 2x$ given that $\sin x = \frac{\sqrt{3}}{4}$ given x is obtuse angled?

☒ A. $-\frac{\sqrt{39}}{8}$

B. $-\frac{\sqrt{3}}{2}$

C. $\frac{\sqrt{3}}{2}$

D. $\frac{\sqrt{39}}{8}$



$$\begin{aligned} \sin 2x &= 2 \sin x \cos x \\ &= 2 \cdot \frac{\sqrt{3}}{4} \cdot -\frac{\sqrt{13}}{4} \end{aligned}$$

- 4 The radius of a spherical balloon is expanding at a constant rate of 1.3 cm s^{-1} . What is the rate of change of the surface area of the balloon when its radius is 6.3 cm ?

- A. $68.61 \text{ cm}^2 \text{ s}^{-1}$
 B. $158.34 \text{ cm}^2 \text{ s}^{-1}$
☒ C. $205.84 \text{ cm}^2 \text{ s}^{-1}$
 D. $498.76 \text{ cm}^2 \text{ s}^{-1}$

$$\frac{dr}{dt} = 1.3$$

$$S = 4\pi r^2$$

$$\frac{dS}{dr} = 8\pi r$$

$$\frac{dS}{dt} = 1.3 \times 8 \times \pi \times 6.3$$

- 5 Using the digits 0, 1, 2, 3, 4 and 5, how many odd numbers between 2000 and 5000 can be formed if no digits are repeated?

- A. 48
☒ B. 96
 C. 120
 D. 360

$$2: 1 \times 4 \times 3 \times 3$$

$$3: 1 \times 4 \times 3 \times 2$$

$$4: 1 \times 4 \times 3 \times 3$$

- 6 Consider the integral:

$$\int_0^k \frac{2}{8 + (2x)^2} dx = \frac{\pi}{8\sqrt{2}}$$

What is the value of k ?

- A. $\frac{1}{\sqrt{3}}$
 B. 1
☒ C. $\sqrt{2}$
 D. $\sqrt{3}$

$$\left[\frac{1}{2\sqrt{2}} \cdot \tan^{-1} \left(\frac{x}{\sqrt{2}} \right) \right]_0^k = \frac{\pi}{8\sqrt{2}}$$

$$\tan^{-1} \left(\frac{k}{\sqrt{2}} \right) - \tan^{-1} 0 = \frac{\pi}{4}$$

$$\tan \frac{\pi}{4} = \frac{k}{\sqrt{2}}$$

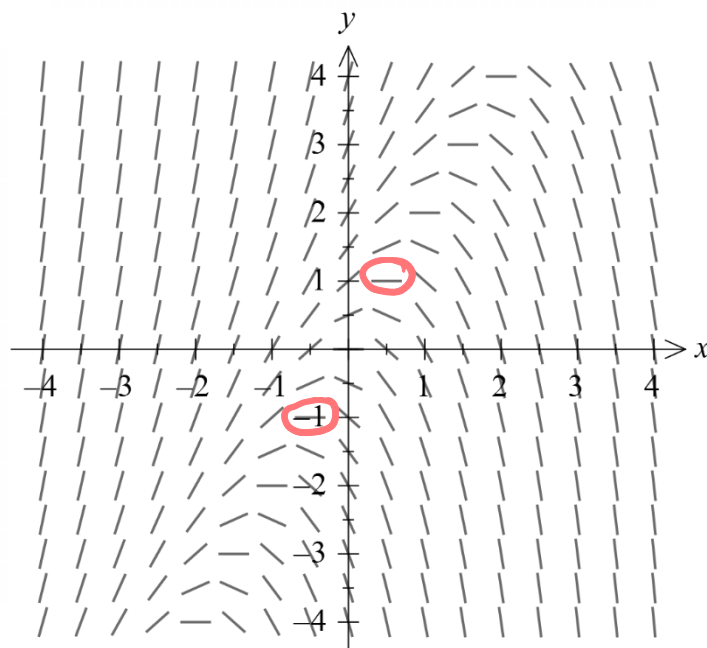
$$1 = \frac{k}{\sqrt{2}}$$

- 7 The polynomial $x^2 + x + 2$ is a factor of the polynomial $x^4 + 4x^2 + x + a$.
What is the value of a ?

A. -6
B. -3
C. 3
D. 6

$$\begin{array}{r}
 x^2 - x + 3 \\
 x^2 + x + 2 \overline{) x^4 + + 4x^2 + x + a} \\
 \underline{x^2 + x^3 + 2x^2 -} \\
 -x^3 + 2x^2 + x \\
 \underline{-x^3 - x^2 - 2x -} \\
 3x^2 + 3x + a \\
 3x^2 + 3x + 6
 \end{array}$$

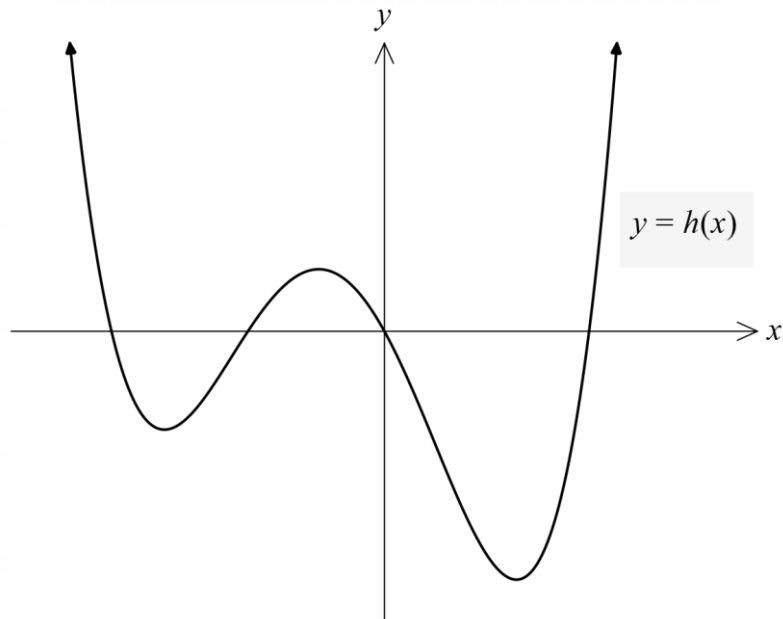
- 8 The direction field for a differential equation is shown below.



Which of the following equations could give this direction field?

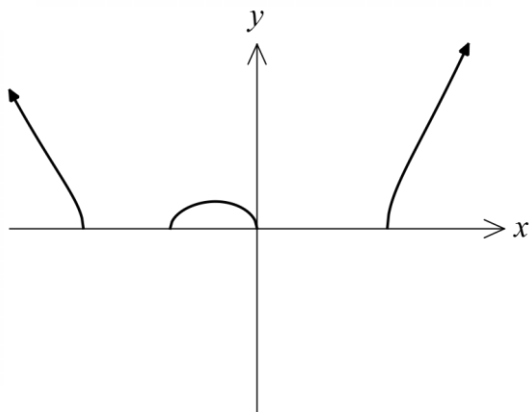
- A. $y' = y - 2x$
B. $y' = y + 2x$
C. $y' = y + x$
D. $y' = x - y$

- 9 The graph of $y = h(x)$ is shown below.

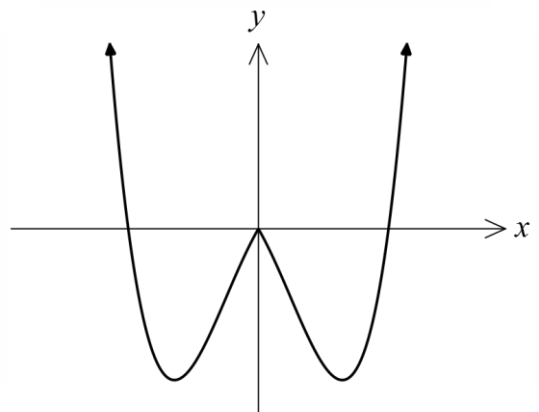


Which graph below shows $y = |h(x)|$?

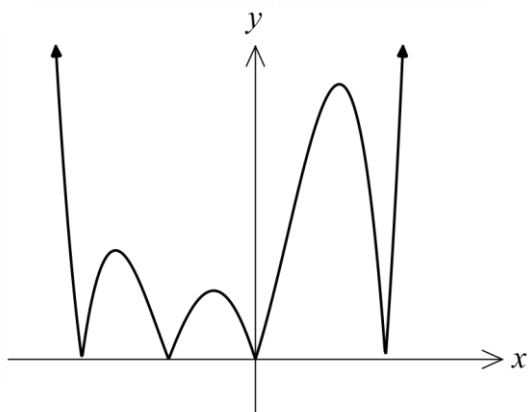
A.



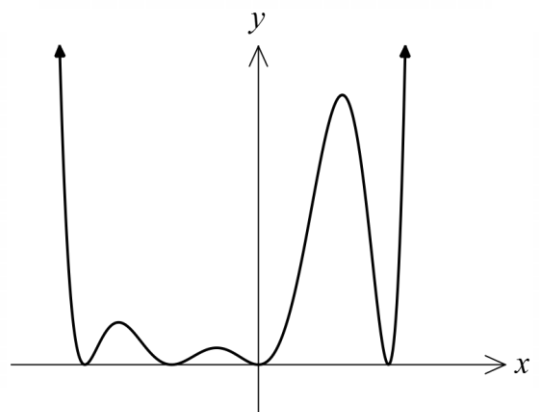
B.



C.



D.



- 10 The graph of the function $y = \sin^{-1}\left(\frac{1}{2}(x-2)\right)$ is to be transformed by a translation by 1 unit to the left, then a horizontal dilation with a scale factor of 2.

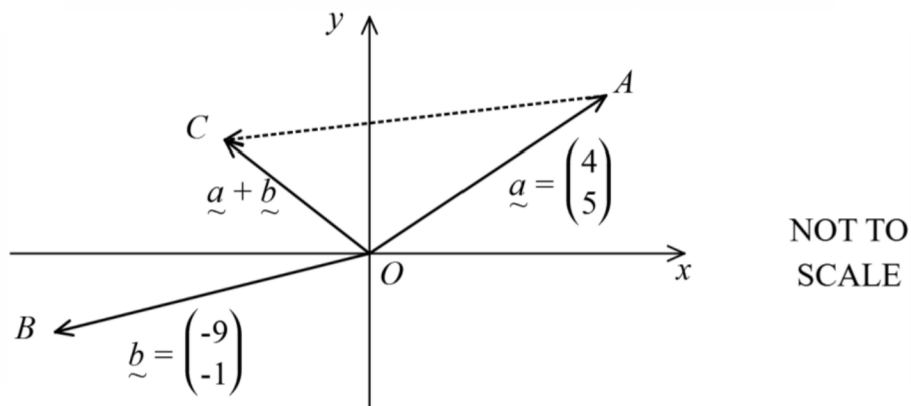
What is the domain of the resultant function?

- A. $-1 \leq x \leq 7$
B. $-2 \leq x \leq 6$
C. $1 \leq x \leq 3$
D. $-\frac{1}{2} \leq x \leq \frac{3}{2}$

$\sin^{-1} x$ -1 to 1
stretch -2 to 2
 $\sin^{-1}\left(\frac{1}{2}(x-2)\right)$ shift 0 to 4
then: -1 to 3
 -2 to 6

Question 11 (16 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows the vectors $\underline{a} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} -9 \\ -1 \end{pmatrix}$ and the resultant vector $\underline{a} + \underline{b}$.



- (i) Show that $\triangle AOC$, formed by the vectors, is an isosceles triangle

2

$$|\underline{a}| = \sqrt{4^2 + 5^2} \\ = \sqrt{41}$$

1 mark: one correct magnitude
OR $\underline{a} + \underline{b}$ correct.

$$\underline{a} + \underline{b} = \begin{pmatrix} -5 \\ 4 \end{pmatrix}$$

$$|\underline{a} + \underline{b}| = \sqrt{(-5)^2 + 4^2} \\ = \sqrt{41}$$

2 marks: all correct including final statement.

$\therefore \triangle AOC$ is isosceles.

- (ii) Find the size of $\angle OAC$.

1

$$\underline{a} \cdot (\underline{a} + \underline{b}) = |\underline{a}| |\underline{a} + \underline{b}| \cos(\hat{AOC})$$

$$-20 + 20 = \sqrt{41} \sqrt{41} \cos \hat{AOC}$$

$$0 = \cos \hat{AOC}$$

$$\therefore \hat{AOC} = 90^\circ$$

$$\therefore \text{since isosceles } \hat{OAC} = 45^\circ$$

①

- (b) (i) Rewrite $\cos x - \sqrt{3}\sin x$ in the form $R\cos(x + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$, $R > 0$.

2

$$R\cos(x + \alpha) = R\cos x \cos \alpha - R\sin x \sin \alpha$$

$$\therefore R = \sqrt{1^2 + (\sqrt{3})^2} \\ = 2$$

$$\tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

$$\therefore \cos x - \sqrt{3}\sin x = 2\cos\left(x + \frac{\pi}{3}\right)$$

①
correct R

①
correct angle

- (ii) Hence, or otherwise, solve $\cos x - \sqrt{3}\sin x = 1$ for $0 \leq x \leq 2\pi$.

2

$$\cos x - \sqrt{3}\sin x = 1$$

$$2\cos\left(x + \frac{\pi}{3}\right) = 1$$

$$0 \leq x \leq 2\pi$$

$$\frac{\pi}{3} \leq x + \frac{\pi}{3} \leq \frac{7\pi}{3}$$

$$\cos\left(x + \frac{\pi}{3}\right) = \frac{1}{2}$$

$$x + \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$$

$$x = 0, \frac{4\pi}{3}, 2\pi$$

✓
✓

① for $\frac{\pi}{3}$

① for all three angles

- (c) Evaluate $\int_1^{\sqrt{2}} \frac{x}{\sqrt{4-x^2}} dx$ using the substitution $u = 4 - x^2$.

3

$$u = 4 - x^2$$

$$du = -2x dx$$

$$x = \sqrt{2}$$

$$u = 2$$

$$x = 1$$

$$u = 3$$

① change bounds

$$\therefore I = -\frac{1}{2} \int_1^{\sqrt{2}} \frac{-2x}{\sqrt{4-x^2}} dx$$

$$= -\frac{1}{2} \int_3^2 \frac{1}{\sqrt{u}} du$$

① correct subst.
must not mix
variables

$$= -\frac{1}{2} [2u^{\frac{1}{2}}]_3^2$$

$$= -\frac{1}{2} (2\sqrt{2} - 2\sqrt{3})$$

$$= \sqrt{3} - \sqrt{2}$$

① answer.

- (d) The coefficient of x in the expansion of $\left(x + \frac{1}{ax^2}\right)^7$ is $\frac{7}{3}$.

3

Find the value(s) of a .

$$T_{n+1} = \binom{7}{n} (x)^7 \left(\frac{1}{ax^2}\right)^n$$

$$= \binom{7}{n} \frac{1}{a^n} \cdot x^{7-n} \cdot x^{-2n}$$

① general term

For x term

$$x^{7-n} \cdot x^{-2n} = x^1$$

$$\therefore 7 - 3n = 1$$

$$3n = 6$$

$$n = 2$$

① correct power

For co-efficient

$$\binom{7}{2} \frac{1}{a^2} = \frac{7}{3}$$

$$\frac{21}{a^2} = \frac{7}{3}$$

$$a^2 = 9$$

$$\therefore a = \pm 3$$

① correct
values of a

(e) Solve the inequality $\frac{1}{x} \leq \frac{1}{x+1}$.

3

$$\frac{1}{x} \leq \frac{1}{x+1} \quad x \neq -1, 0$$

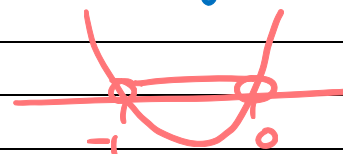
$$x(x+1)^2 \leq x^2(x+1)$$

① for attempt at
to solve - critical
- square
- graph.

$$x(x+1)^2 - x^2(x+1) \leq 0$$

$$x(x+1)(x+1-x) \leq 0$$

$$x(x+1) \leq 0$$



$$\therefore -1 < x < 0$$

① for correct inequality signs
① for correct values.

15
Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Use mathematical induction to show that $9^{n+2} - 4^n$ is divisible by 5 for all positive integers n .

4

For $n=1$

$$9^{1+2} - 4^1$$

$$= 9^3 - 4$$

$$= 729 - 4$$

$$= 725 \text{ which is divisible by 5}$$

$\therefore$ The statement is true for $n=1$

Assume true for $n=k$, $k \in \mathbb{Z}^+$

$$9^{k+2} - 4^k = 5M \quad M \in \mathbb{Z}$$

$$4^k = 9^{k+2} - 5M$$

Prove true for $n=k+1$

$$9^{(k+1)+2} - 4^{k+1}$$

$$= 9^{k+3} - 4 \cdot 4^k$$

$$= 9^{k+3} - 4(9^{k+2} - 5M)$$

$$= 9 \cdot 9^{k+2} - 4 \cdot 9^{k+2} + 20M$$

$$= 5 \cdot 9^{k+2} + 20M$$

$$= 5(9^{k+2} + 4M)$$

from assumption

① use of assumption

① correct solution

which is divisible by 5 as $(9^{k+2} + 4M)$ is an integer.

$\therefore$ When the statement is true for $n=k$ it is true for $n=k+1$

$\therefore$ since the statement is true for $n=1$ it is true for $n=2$ & so on for all positive integers.

- (b) The polynomial $P(x) = x^4 - 3x^3 + ax^2 + bx - 6$ leaves a remainder of 8 when divided by $(x + 1)$. If $(x - 3)$ is a factor of $P(x)$, find the values of a and b .

$$P(-1) = 8 = (-1)^4 - 3(-1)^3 + a(-1)^2 + b(-1) - 6$$

$$8 = -2 + a - b$$

$$10 = a - b$$

$$b = a - 10 \quad (1)$$

① use of remainder theorem

$$P(3) = 0 = (3)^4 - 3(3)^3 + a(3)^2 + 3b - 6$$

$$0 = -6 + 9a + 3b$$

$$6 = 9a + 3b$$

$$2 = 3a + b$$

① use of factor theorem

subst ① into ②

$$2 = 3a + a - 10$$

$$4a = 12$$

$$a = 3$$

① answer

$$b = 3 - 10$$

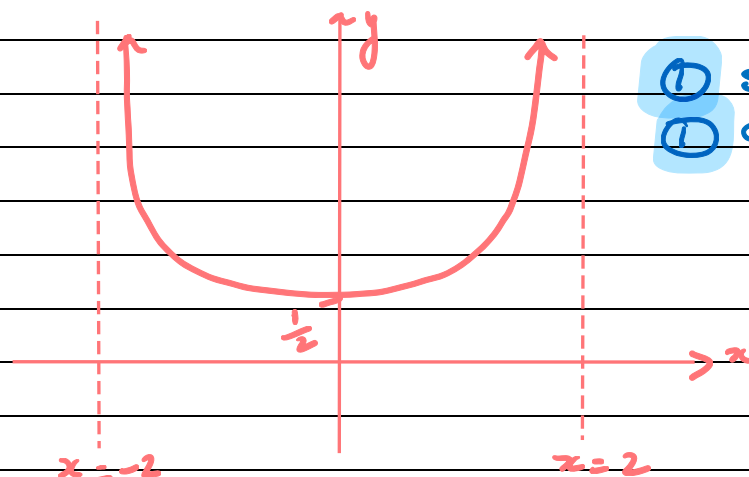
$$b = -7$$

① answer.

(c) Consider the function $f(x) = \sqrt{4-x^2}$. A new function is created such that $g(x) = \frac{1}{f(x)}$.

- (i) Sketch $g(x)$, showing all important features.
(You do NOT need to use calculus to verify any points on the graph.)

$$g(x) = \frac{1}{\sqrt{4-x^2}}$$



① shape
① asymptotes and y-intercept

- (ii) Find the exact area bound by $g(x)$, $x = \sqrt{2}$, $x = \sqrt{3}$ and the x -axis.

$$A = \int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$$

$$= \left[\sin^{-1}\left(\frac{x}{2}\right) \right]_{\sqrt{2}}^{\sqrt{3}}$$

$$= \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \sin^{-1}\left(\frac{\sqrt{2}}{2}\right)$$

$$= \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$$

①

①

- (d) Given that ${}^nC_4 = p$ and ${}^nC_5 = q$, use the identities of Pascal's Triangle, or otherwise, to find a simplified expression for ${}^{n+1}C_{\cancel{n-5}} + {}^nC_{n-4}$ in terms p and q .

2

$${}^{n+1}C_5 = {}^nC_4 + {}^nC_5$$

$$= p + q$$

$${}^{n+1}C_{(n+1-5)} = {}^{n+1}C_5 = {}^{n+1}C_{n-4} = p + q$$

$${}^nC_{n-4} = {}^nC_4 = p$$

① for one identity

$$\therefore {}^{n+1}C_{n-5} + {}^nC_{n-4} = p + q + p$$

$$= 2p + q$$

① correct

ERROR:

$${}^nC_{n-4} = {}^nC_4 = p$$

① still possible.

$${}^nC_5 = \frac{n!}{5!(n-5)!} = q$$

$${}^{n+1}C_{n-5} = \frac{(n+1)!}{(n-5)!(n+1-(n-5))!}$$

①

$$= \frac{(n+1)!}{6!(n-5)!}$$

$$= \frac{(n+1) \cdot n!}{6 \times 5! \cdot (n-5)!}$$

$$= \left(\frac{n+1}{6}\right)q$$

$$\therefore {}^{n+1}C_{n-5} + {}^nC_{n-4} = \left(\frac{n+1}{6}\right)q + p$$

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) The letters of the word PERSEVERE are arranged in a row.

(i) How many different arrangements of the letters are possible?

2

$$\text{No.} = \frac{9!}{4! 2!}$$

① for 9! and a division

$$= 7560$$

① correct answer

(ii) What is the probability that an arrangement will have all E's together in one group and all the R's together in one group?

1 2

$$\text{No. with E + R} = 5! = 120$$

P E R S V

$$\therefore P = \frac{120}{7560}$$

$$= \frac{1}{63}$$

① for answer

(b) If $2\sin\left(\theta - \frac{\pi}{3}\right) = \cos\left(\theta - \frac{\pi}{3}\right)$ express $\tan\theta$ in the form $a + b\sqrt{c}$, where a , b , and c are integers.

3

$$2\sin\left(\theta - \frac{\pi}{3}\right) = \cos\left(\theta - \frac{\pi}{3}\right)$$

$$\therefore 2\tan\left(\theta - \frac{\pi}{3}\right) = 1$$

① use of one trig identity

$$2 \cdot \frac{\tan\theta - \tan\frac{\pi}{3}}{1 + \tan\theta \tan\frac{\pi}{3}} = 1$$

① use of sum

$$2\tan\theta - 2\sqrt{3} = 1 + \sqrt{3}\tan\theta$$

$$2\tan\theta - \sqrt{3}\tan\theta = 1 + 2\sqrt{3}$$

$$\tan\theta = \frac{1 + 2\sqrt{3}}{2 - \sqrt{3}}$$

$$= 8 + 5\sqrt{3}$$

① correct.

(c) Consider $f(x) = 4\sin^2\left[2\left(x + \frac{\pi}{8}\right)\right] - 2$, $0 < x < \frac{\pi}{4}$.

(i) Verify that $f(x)$ could be a probability density function.

3

For a pdf $A = 1$

$$4 \sin^2\left(2\left(x + \frac{\pi}{8}\right)\right) = 4\left(\frac{1}{2}\left(1 - \cos\left(4\left(x + \frac{\pi}{8}\right)\right)\right)\right) \\ = 2 - 2\cos\left(4x + \frac{\pi}{2}\right) \quad (1)$$

$$\therefore A = \int_0^{\pi/4} 2 - 2\cos\left(4x + \frac{\pi}{2}\right) - 2 \, dx$$

$$= \int_0^{\pi/4} -2\cos\left(4x + \frac{\pi}{2}\right) \, dx$$

$$= \left[-\frac{1}{2}\sin\left(4x + \frac{\pi}{2}\right)\right]_0^{\pi/4} \quad (1)$$

$$= -\frac{1}{2}\sin\left(\frac{3\pi}{2}\right) - \left(-\frac{1}{2}\sin\left(\frac{\pi}{2}\right)\right)$$

$$= \left(-\frac{1}{2} \cdot -1\right) - \left(-\frac{1}{2} \cdot 1\right)$$

$$= \frac{1}{2} + \frac{1}{2} = 1 \quad \therefore f(x) \text{ is a pdf.} \quad (1)$$

(ii) Find the mode of the distribution.

2

Max occurs when

$$\sin^2\left(2x + \frac{\pi}{4}\right) = 1$$

$$\sin\left(2x + \frac{\pi}{4}\right) = 1 \quad \text{or} \quad \sin\left(2x + \frac{\pi}{4}\right) = -1 \quad (1)$$

$$2x + \frac{\pi}{4} = \frac{\pi}{2} \\ 2x = \frac{\pi}{4}$$

$$x = \frac{\pi}{8} \quad (1)$$

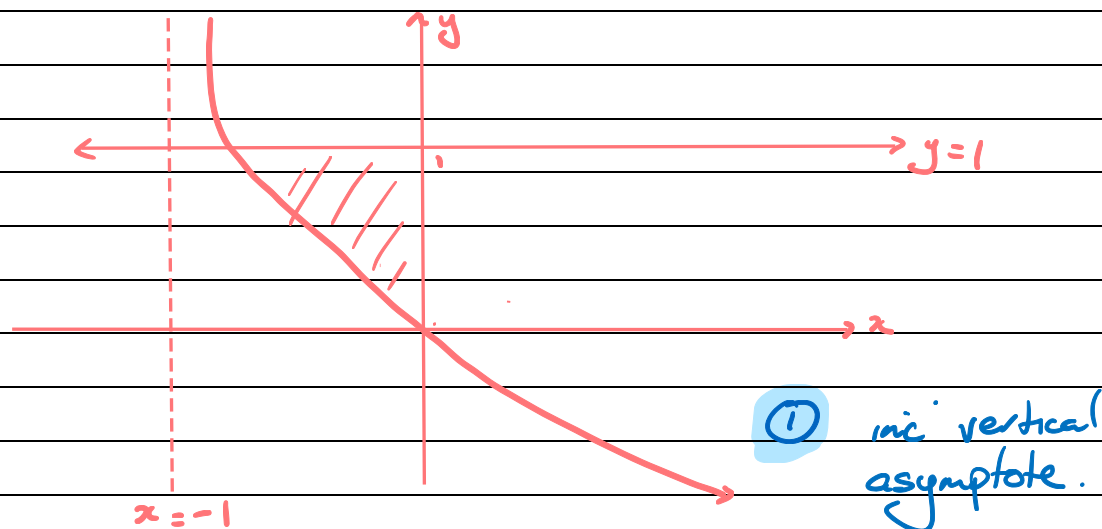
$$2x + \frac{\pi}{4} = \frac{3\pi}{2} \\ 2x = \frac{5\pi}{4}$$

$x = \frac{5\pi}{8}$
outside domain

- (d) The curve $y = -\log_e(x+1)$, the y -axis and the line $y = 1$ enclose a region in the number plane.

- (i) Sketch the region, clearly showing any asymptotes and intercepts with axes.

1



- (ii) The region is rotated about the y -axis to form a solid of revolution. Find the volume, giving your answer correct to 3 significant figures.

3

$$y = -\ln(x+1)$$

$$-y = \ln(x+1)$$

$$x+1 = e^{-y}$$

$$x = e^{-y} - 1$$

①

$$x^2 = (e^{-y} - 1)^2$$

$$= e^{-2y} - 2e^{-y} + 1$$

$$V = \pi \int_0^1 x^2 dy$$

$$= \pi \int_0^1 e^{-2y} - 2e^{-y} + 1 dy$$

$$= \pi \left[-\frac{1}{2} e^{-2y} + 2e^{-y} + y \right]_0^1$$

①

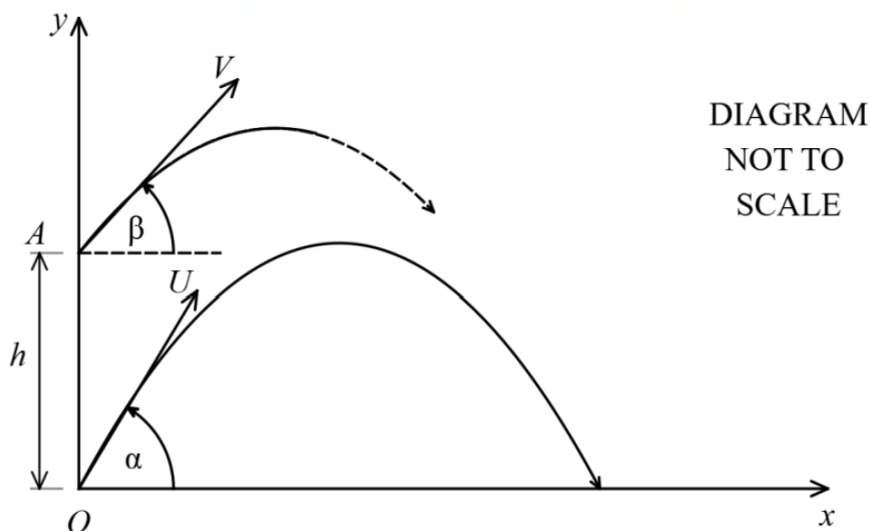
$$= \pi \left(-\frac{1}{2} e^{-2} + 2e^{-1} + 1 - \left(-\frac{1}{2} + 2 \right) \right)$$

$$= 0.528074207 = 0.528 \text{ to 3 s.f.}$$

①

Question 14 (14 marks) Use a SEPARATE writing booklet.

- (a) An object is projected on level ground from the origin, O , with initial speed U m/s at an angle of elevation of α . At the same instant another object is projected from a point A which is h m above the origin. The second object is projected with initial speed V m/s at an angle of elevation of β , where $\beta < \alpha$.



- (i) The parametric equations of displacement for the object projected from the origin, O , are

$$x = Ut \cos \alpha \text{ and } y = Ut \sin \alpha - \frac{1}{2}gt^2$$

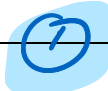
(DO NOT PROVE THESE)

Write down the two parametric equations of displacement for the object projected from A .

From A:

$$x = Vt \cos \beta$$

$$y = Vt \sin \beta - \frac{1}{2}gt^2 + h$$



- (ii) If the objects collide T seconds after they are projected, show that

$$T = \frac{h \cos \beta}{U \sin(\alpha - \beta)}.$$

When collide at $t = T$, both x & y are equal.

For x :

$$VT \cos \beta = UT \cos \alpha$$

$$\therefore V = \frac{U \cos \alpha}{\cos \beta}$$

① for equating x or y

①

For y :

$$VT \sin \beta - \frac{1}{2} g T^2 + h = UT \sin \alpha - \frac{1}{2} g T^2$$

$$h = UT \sin \alpha - VT \sin \beta$$

①

$$h = T(U \sin \alpha - V \sin \beta)$$

$$= T \left(U \sin \alpha - \frac{U \cos \alpha \sin \beta}{\cos \beta} \right)$$

$$h = UT \left(\frac{\sin \alpha \cos \beta - \cos \alpha \sin \beta}{\cos \beta} \right)$$

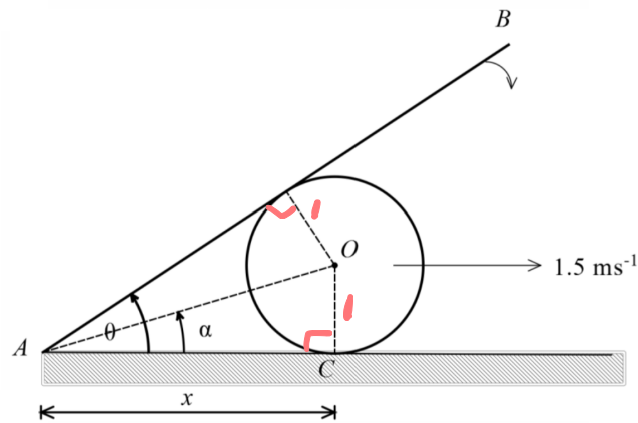
①

$$h = UT \frac{\sin(\alpha - \beta)}{\cos \beta}$$

$$\frac{h \cos \beta}{U \sin(\alpha - \beta)} = T$$

Q.E.D

Q 14(b)



- (i) Write a relationship between $\angle OAC$, OC and AC .

1

$$\tan \alpha = \frac{1}{x} \quad \text{or} \quad \tan \alpha = \frac{OC}{AC}$$

①

- (ii) Hence, or otherwise, find the rate of change of θ in radians per second when $x = 4$ m, giving your answer correct 3 decimal places.

4

$$\theta = 2\alpha \quad (\text{bisected angle in a kite})$$

$$\frac{d\theta}{dt} = \frac{dx}{dt} \times \frac{d\theta}{dx}$$

$$\frac{dx}{dt} = 1.5 \text{ m/s}$$

$$\tan \alpha = \frac{1}{x}$$

$$\alpha = \tan^{-1}\left(\frac{1}{x}\right)$$

$$2\alpha = 2 \tan^{-1}\left(\frac{1}{x}\right)$$

$$\therefore \theta = 2 \tan^{-1}(x^{-1})$$

① expression for θ

$$\frac{d\theta}{dx} = \frac{-2x^{-2}}{1+(x^{-1})^2}$$

$$= \frac{-2}{x^2+1}$$

$$\text{① } \frac{dx}{d\theta}$$

$$\therefore \frac{d\theta}{dt} = 1.5 \times \frac{-2}{x^2+1}$$

① attempt at related rate

$$\text{at } x = 4$$

$$\frac{d\theta}{dt} = 1.5 \times \frac{-2}{4^2+1}$$

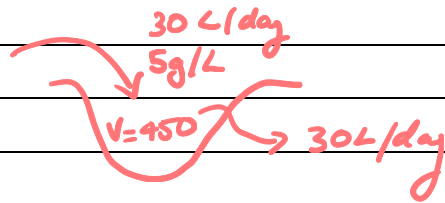
① for answer

$$= -0.176 \text{ rads/sec}$$

Q14(c)

- (i) Show that a differential equation to model the mass of pollutant after t days is

$$\frac{dm}{dt} = 150 - \frac{m}{15}$$



$$\frac{dm}{dt} = \text{rate in} - \text{rate out}$$

$$= \frac{5g}{L} \times \frac{30L}{\text{day}} - \frac{m}{450L} \times \frac{30L}{\text{day}}$$

$$\frac{dm}{dt} = 150 - \frac{m}{15}$$

①

- (ii) What is the concentration of the pollutant in the pond after 10 days?

$$\frac{dm}{dt} = 150 - \frac{1}{15} m$$

$$\frac{dm}{dt} = \frac{2250 - m}{15}$$

$$\frac{1}{2250 - m} dm = \frac{1}{15} dt$$

$$\int \frac{1}{2250 - m} dm = \int \frac{1}{15} dt$$

$$-\ln|2250 - m| = \frac{t}{15} + C$$

at $t=0$ $m=0$

$$-\ln|2250| = 0 + C$$

$$C = -\ln|2250|$$

① correct integral

$$\therefore -\ln|2250 - m| = \frac{t}{15} - \ln|2250|$$

at $t=10$

$$-\ln |2250 - m| = \frac{10}{15} - \ln |2250|$$

$$\ln |2250 - m| = \ln 2250 - \frac{2}{3}$$

$$2250 - m = e^{\ln 2250 - \frac{2}{3}}$$

$$2250 - m = 2250 \cdot e^{-\frac{2}{3}}$$

$$\begin{aligned} m &= 2250 - 2250 \cdot e^{-\frac{2}{3}} \\ &= 1094.811 \text{ g. } \textcircled{1} \text{ correct } m. \end{aligned}$$

$$\begin{aligned} \therefore \text{Concentration} &= \frac{1094.811}{450} \\ &= 2.4329... \\ &= 2.43 \text{ g/L. } \textcircled{1} \text{ correct } C. \end{aligned}$$